

Arrays - VII

Tuesday, 5 May 2026 10:01 PM

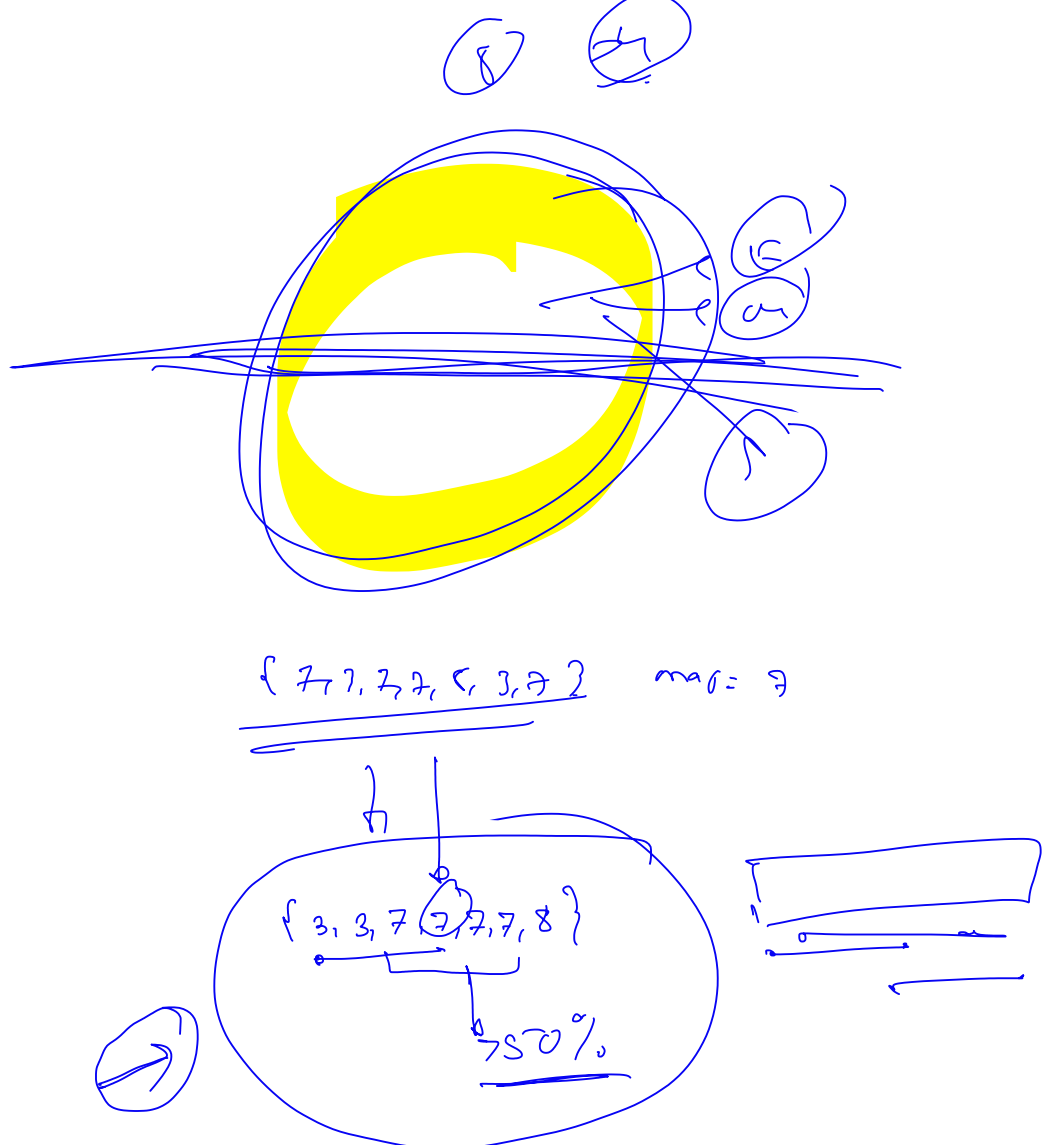
Q.1) Majority Element

IP = { 7, 0, 0, 1, 7, 8, 3, 2, 8, 8, 8, 8, 8, 8, 8 }

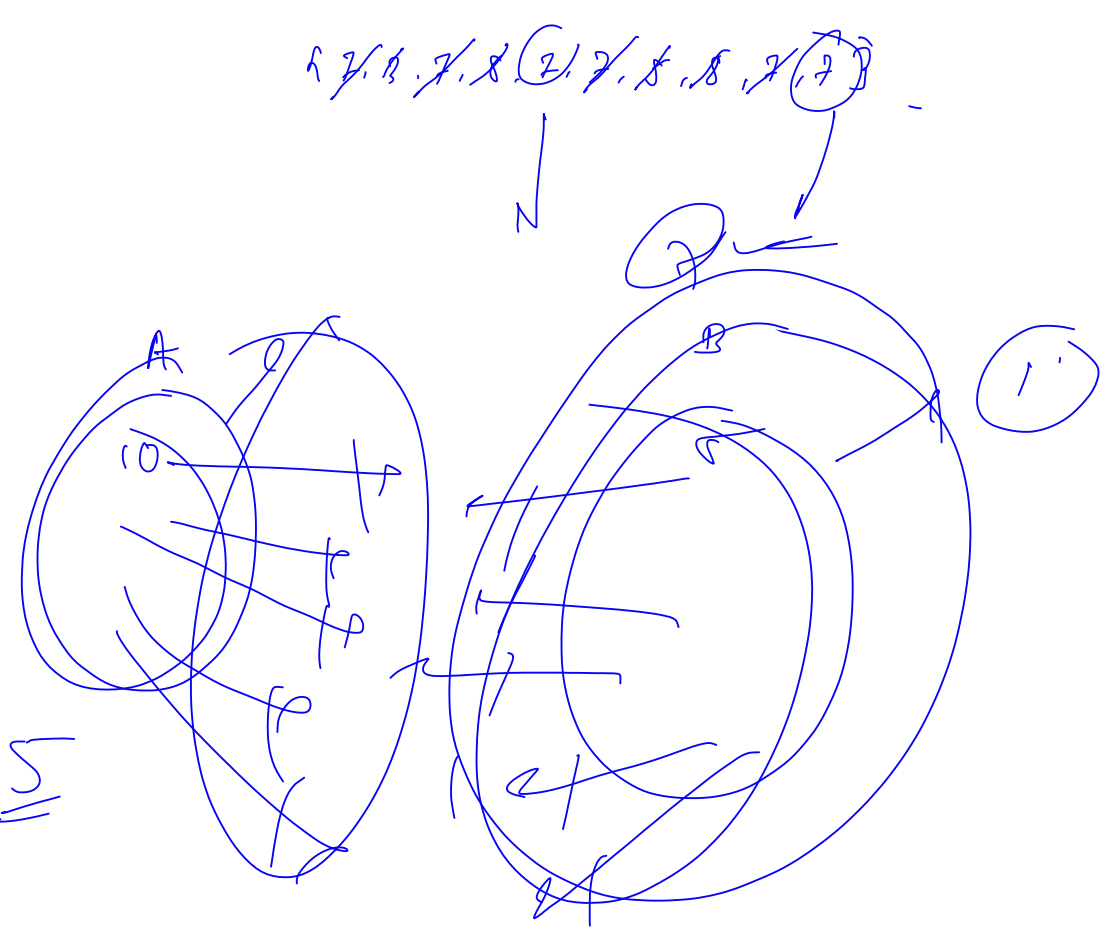
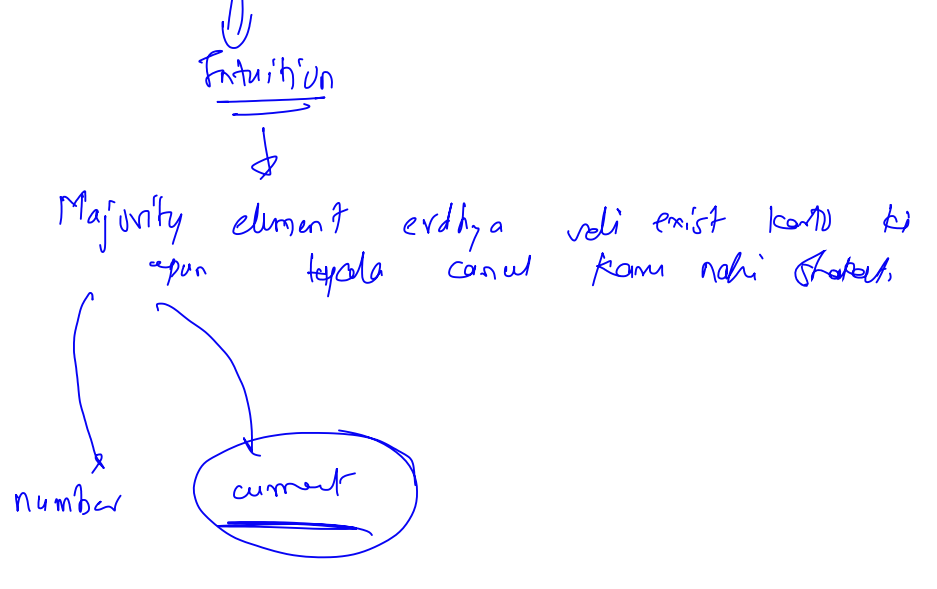
Ques - majority element exist

O/P - 8

IP = { 1, 1, 2, 2, 3, 3, 5, 5 }



- Approach 1 - 2 for loops
- Approach 2 - frequency array.
- Approach 3 - HashMap
update - $rf()$
reset -
- Approach 4 - Sorting - mid element.
- Approach 5 - Moore's Voting Algorithm.



Dry Run -

arr = { 2, 2, 1, 1, 2, 2, 3 }

candidate = 2, count = 1

candidate = 2, count = 2

candidate = 2, count = 1

candidate = 1, count = 0

candidate = 1, count = 1

candidate = 1, count = 0

candidate = 2, count = 1

2

Code -

```
public static int majorityElement (int[] nums) {  
    int candidate = 0;  
    int count = 0;  
    for (int i = 0; i < nums.length; i++) {  
        if (count == 0) {  
            candidate = nums[i];  
        }  
        if (nums[i] == candidate) {  
            count++;  
        }  
        else {  
            count--;  
        }  
    }  
    return candidate;  
}
```

Q. Majority Element $(N/2) \Rightarrow (N/3)$